S^n, ~~$S^1 \times S^1$~~, $\mathbb{RP}^n$,

$(n \geq 2)$
$\downarrow$
$\pi_1 = 0$.

2-dim mfd

$L(p,q)$

$M \# N$. $\wedge$ $\vee$

$D^1 \times S^1 / e^{2\pi i p}, e^{2\pi i q}$

$[0,1]^2$

a b a b

torus.

$\mathbb{Z} \times \mathbb{Z}$

Klein.

Klein. free product.

ababab$b$..

$aba^{-1}b^{-1} = e$.

$\langle a, b \mid aba = b \rangle$.

$\langle a, b \mid ab = ba \rangle$
$\mathbb{Z} \times \mathbb{Z}$

$ab = ba \to$ group

$S^1 \vee S^1$

$S^1 \vee S^2$

$S^1 \vee S^1$
$\mathbb{Z} * \mathbb{Z}$

$T: \mathbb{Z} \times \mathbb{Z}$

T: $\mathbb{Z} * \mathbb{Z}$

K: $\mathbb{Z} * \mathbb{Z}$

theorem

Theorem

$S^1 \vee S^1$
$\mathbb{Z} * \mathbb{Z}$

S^1
$\updownarrow$
$\mathbb{Z}$

$S^3 : 0$.
$\downarrow$
$\mathbb{R}^3 : 0$.

M: n-dim
with $n \geq 3$

then $\pi_1(M) \cong \pi_1(M \setminus \circ)$.
$\cong \pi_1(M \setminus 2pt)$. $\pi_1(S^n) = 0$
$n \geq 2$.
$\cong \pi_1(M \setminus \{x_i\})$.

$\pi_1(\mathbb{RP}^n) = \mathbb{Z}_2$
$(n \geq 2)$.

tree

$S^1 \vee S^1 \vee \mathbb{R}$.
$\downarrow$
$\mathbb{Z} * \mathbb{Z}$

$\mathbb{R}$

$: S^1 \vee S^1 \vee S^1$
$\mathbb{Z} * \mathbb{Z} * \mathbb{Z}$

Example.

N

$S^2 / N = S$ N

$\to \pi_1(\cdots)$

$S^1 \vee S^2$.

S^1: 0-cell 1개
1-cell 1개.

S^2: 0-cell 1개
2-cell 1개

$S^1 \vee S^1$

$\mathbb{RP}^2$: 0-cell : 1개
1-cell : 1개
2-cell 1개

a b a^{-1} b^{-1}

$aba^{-1}b^{-1} = 0$.

$\langle a, b \mid ab = b a^{-1} \rangle$

$\mathbb{Z} * \mathbb{Z} / \alpha$

$\mathbb{RP}^2 : \mathbb{Z}_2$ $\langle a \mid a^2 \rangle$
$a^2 = 1$

a a

$\mathbb{RP}^3$: ~~0-cell 1개~~
~~1-cell 1개~~
~~2-cell 1개~~
~~3-cell 1개~~

$\pi_1(\mathbb{RP}^3) = \pi_1(\mathbb{RP}^2) = \mathbb{Z}_2$.

~~$\pi_1(\ldots)$~~

$x^2 + y^2 + z^2 = 1$
$\cup$ $x^2 + y^2 \leq 1$
$\cup$ $(0,0,z)$ $-1 \leq z \leq 1$

$(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$

$\mathbb{R}/\mathbb{Z}$

S^1

?

$Z = x^4 + y^4$
$\downarrow$ exp

$p = 0$

$\frac{1}{p_n + z}$.

$K_1 = f(4, z)$
$K_2 = f_4 \ldots$

$\int \frac{R dz}{y}$ $\int \frac{1}{z} dz$
$\frac{4}{2\pi i}$

$2\pi \cdot X(C^{-1})$
$\downarrow$
0